



Teachers' Implementation of the *Identity Project* Is Associated With Increases in U.S. High School Students' Ethnic-Racial Identity Exploration

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Abstract

Ethnic-racial identity formation has significant consequences for positive youth development. Existing findings support the efficacy of the *Identity Project*, a school-based ethnic-racial identity intervention, when delivered by researchers; however, effectiveness of the program when delivered by teachers is unknown. This study examined changes in adolescents' ($N = 180$; 42.2% male, 50.6% female, 6.7% another gender identity; $M_{\text{age}} = 14.11$, $SD = 0.33$; 38.3% Latinx, 33.9% White, 15.0% Black, 9.4% Asian American, 3.3% another ethnoracial background) ethnic-racial identity exploration as a function of their teachers' implementation of the *Identity Project*. Findings indicated that ethnic-racial identity exploration significantly increased from pretest to posttest, and this did not vary based on familial ethnic-racial socialization, student-teacher ethnoracial match/mismatch, gender, immigrant status, or ethnoracial background. This study provides preliminary evidence that U.S. educators can be trained to efficaciously implement the *Identity Project* with high school students and, furthermore, that this approach to program dissemination may not only facilitate scale-up but also result in greater gains for adolescents relative to research-led implementations.

Keywords Intervention · Ethnic-racial identity · Teachers · School

Introduction

Identity formation is a key developmental task of adolescence (Meeus, 2011) and can have significant implications for adjustment (Benson et al., 2006). Furthermore, individuals' identities are multifaceted and informed by their various social group memberships (Umaña-Taylor, 2011). Due to the ubiquitous nature of race and ethnicity in U.S. society, ethnic-racial¹

identity is one such social group membership that is important for the development and adjustment of U.S. adolescents (Umaña-Taylor, 2023a). Indeed, research has demonstrated more positive psychosocial outcomes among adolescents who have a clearer sense of the role of race and ethnicity in their lives, and who have achieved this clarity through a process of exploration of their ethnoracial backgrounds (Umaña-Taylor & Rivas-Drake, 2021). Given these positive links, the *Identity Project* intervention (Umaña-Taylor, 2018) was designed to promote the developmental components of ethnic-racial identity exploration and resolution. Efficacy testing has provided support for the *Identity Project* when delivered by researchers in the U.S. (Umaña-Taylor, Douglass et al., 2018) and in Italy (Cecon et al., 2023); however, the effectiveness of the program when delivered by teachers is unknown. Teachers are key socialization agents during adolescence (Wang & Eccles, 2012), have a significant influence on students' psychosocial (Karataş et al.,

¹ The term ethnic-racial is used throughout this article to acknowledge the intersecting nature of ethnicity and race in the psychological process of forming an identity based on individuals' sometimes-overlapping lived experiences that are shaped by their ethnic and racial group memberships (Umaña-Taylor et al., 2014). In addition, the term ethnic-racial is used to acknowledge the experiences of individuals belonging to social identity groups that have been socially constructed as 'ethnic' (i.e., Latinx/Hispanic; Humes et al., 2011), but whose lived experiences reflect a racialized othering effect (Massey, 2014).

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2023) and academic outcomes (Aronson & Laughter, 2016), and are explicitly trained in effectively teaching the age groups they serve. When trained to implement a curriculum such as the *Identity Project*, teachers' increased content knowledge and self-reflection on issues of ethnicity, race, and identity can help them strengthen their relationships with students and better connect students' identities to the course content (Nojan, 2020). Together with their advanced pedagogical skills, these factors may make teachers particularly effective deliverers of the *Identity Project*. As such, the current study examined whether students' ethnic-racial identity exploration increased from the time before their teachers began implementing the *Identity Project* in their classrooms to shortly after their teachers completed the last *Identity Project* lesson.

Overview of the *Identity Project* Curriculum and its Effects on ERI Exploration

Grounded in developmental (Erikson, 1968) and social psychological (Tajfel & Turner, 1986) theories, the *Identity Project* is a universal health promotion intervention that targets the developmental competency of ethnic-racial identity formation. It was explicitly designed to be implemented during adolescence to leverage the significant changes in cognition and social development that take place during this developmental period (Steinberg, 2005), which increase individuals' ability to think about abstract constructs, sharpen their metacognitive and social perspective taking abilities, and increase their awareness of, exposure to, and involvement in ethnoracially diverse spaces that can lead to greater salience of ethnic-racial identity and prompt exploration of and reflection about one's identity. As such, the specific intervention targets of the *Identity Project* are ethnic-racial identity exploration and, subsequently, ethnic-racial identity resolution (Umaña-Taylor, 2018). According to the theory of intervention for the *Identity Project*, as individuals engage in greater exploration of their ethnoracial backgrounds, this will promote increased reflection on the personal meaning of their ethnic-racial identity for their sense of self; as a result of engaging in this process, they will feel a greater sense of global identity cohesion, they will have less identity confusion, and these experiences will promote more positive psychosocial adaptation (Umaña-Taylor & Douglass, 2017).

The *Identity Project* includes eight sessions, lasting approximately 60 min each, and delivered once per week (see Table 1). It was designed to be implemented by one facilitator and in peer-based settings, ideally in school as part of the standard curriculum, to leverage the importance of peer relationships during adolescence, avoid selection bias, and facilitate scale-up (Umaña-Taylor & Douglass, 2017). The *Identity Project* curriculum provides students with strategies, tools, and the time with which to discuss

Table 1 Overview of *Identity Project* Sessions

Session 1: Unpacking Identity Session 1 introduces students to the concepts of race and ethnicity. The concept of identity is discussed, acknowledging the multiple identities that individuals hold and their importance in students' lives. Activities clarify that everyone has an ethnic-racial identity. The "Identity Backpack" activity is used to give students a concrete way to think about all of the things that make up their identity, and how those things can change over time and across situations.

Session 2: Group Differences - Within and Between Session 2 focuses on the distinction between differences "within" groups versus differences "between" groups. The underlying premise is that how we categorize and label groups of people is socially constructed. Students spend time examining stereotypes about group membership and distinguishing themselves and their identities from those stereotypes they identify.

Session 3: Stories of our Past Session 3 expands students' understanding of how and where discrimination has taken place over the course of history in the United States. The purpose of the session is to engage students in a discussion regarding historical and contemporary experiences of discrimination and the significant consequences of these experiences.

Session 4: Symbols, Traditions, and Rites of Passage Session 4 is focused on increasing students' exploration and understanding of symbols, traditions, rites of passage, and/or rituals for one or multiple ethnic-racial heritages with which they identify. Students also begin the process of identifying influential parental, grandparental, community or other extended family figures in their lives and how they may (or not) play a role in influencing the ways in which students think about their own identity.

Session 5: My Family Mapping In Session 5, students practice strategies and tools with which to explore their ethnic-racial and cultural heritages by gathering information about their family members (broadly and inclusively defined by students), considering potential influences across generations, and building their understanding of how family members can have different degrees of influence in their lives. Session 5 also focuses on increasing students' awareness of the vast diversity that exists in family compositions and histories.

Session 6: Photo Processing and Storyboards Session 6 is designed to increase students' sense of clarity regarding the meaning that various ethnic-racial symbols have for them and their identities. Students research and consider the personal meanings of photographs they take and what these images symbolize. They use these to create a storyboard that represents their ethnic-racial identities.

Session 7: Ethnic-Racial Identity as a Journey Session 7 promotes the idea of "ethnic-racial identity as a journey," meaning that the way that we think and feel about our ethnic-racial identity changes over the course of our lifetime. This session is also designed to engage students in conversations and reflection regarding the notion that different people feel different ways about their ethnic-racial identity, and there is no single or "correct" way to identify with one's ethnic-racial identity.

Session 8: Grand Finale – Celebrations and Closing Session 8 serves as a celebration of students' work during the *Identity Project*. The session begins with a general review of themes, and then moves into an informal setting during which visitors can explore what the students have produced during the *Identity Project*. The session also includes brief presentations by students in which they share food, drinks, artifacts, music, and/or other entertainment related to the heritages they have identified throughout the course of the sessions.

Sessions last approximately 60 min each and are designed to be delivered once per week.

race, ethnicity, and identity in the classroom setting, explore their own ethnic-racial identities, learn about their peers' and teachers' ethnic-racial identities, and reflect on the meaning of this aspect of their identity for their sense of self. Moreover, the manualized curriculum enables educators to have planned activities and follow a learning arc in which students are introduced to and given the opportunity to share their knowledge about basic concepts related to race and ethnicity (e.g., cultural traditions, ethnoracial stereotypes), explore their identities in relation to these concepts, have opportunities to think about themselves and

others in the context of these concepts, and ultimately feel a sense of agency in deciding who they are and how they will think of themselves with respect to race and ethnicity (see Umaña-Taylor, 2018).

The efficacy of the *Identity Project* was tested in a randomized controlled trial (see Umaña-Taylor, Douglass et al., 2018) in which eight classrooms were randomized to either receive the *Identity Project* curriculum (i.e., intervention group) or the *Academic Success* curriculum (i.e., control group), which focused on learning about and exploring different training and educational opportunities after high school (e.g., military, vocational training, community college, 4-year universities). Both curricula were developed and taught by the same facilitators (the PI and a postdoctoral fellow each taught both curricula) and students experienced the same amount of contact time and pedagogical strategies in both curricula (e.g., if a homework assignment was given in the *Identity Project* on a particular day, a homework assignment was also given in the *Academic Success* curriculum on the same day) – essentially, the only difference between the two conditions was the content of the lessons. Findings provided support for the efficacy of the *Identity Project* curriculum, with students in the *Identity Project* demonstrating significant increases in ethnic-racial identity exploration at the 12-week posttest², which subsequently led to significant increases in ethnic-racial identity resolution at the 18-week posttest, compared to students in the control condition (Umaña-Taylor, Douglass et al., 2018). Consistent with the theory of intervention, follow-up analyses approximately one year later demonstrated that these increases in ethnic-racial identity exploration and resolution attributable to the *Identity Project* (compared to control) were subsequently associated with higher self-esteem, lower depressive symptoms, greater global identity cohesion, and better grades; there was also evidence of positive impacts on intergroup contact attitudes and students' academic engagement (Umaña-Taylor, Kornienko et al., 2018).

Findings from the efficacy trial were promising. However, given that the *Identity Project* was developed with the intention of wide dissemination in school-based settings, an important follow-up question was whether support for the program's efficacy would hold when implemented by teachers – an element of intervention *effectiveness*. From a prevention science framework, effectiveness trials can provide evidence regarding the external validity of an intervention by exploring whether the same outcomes can be

achieved when the intervention is delivered under conditions that approximate real world settings, such as when the intervention is delivered by the typical service provider in the context of interest (Marsiglia et al., 2017). With respect to the *Identity Project*, the key question of interest was, “Will the *Identity Project* promote increases in ethnic-racial identity exploration (i.e., the most proximal intervention target) when delivered by classroom teachers?” Although the current study is not an effectiveness trial, as there is not a control group, it enables an initial examination of whether students demonstrate changes in ethnic-racial identity exploration when their classroom teachers implement the curriculum. Furthermore, findings from the current study can be evaluated in relation to those from the original efficacy trial to further build an understanding of how students' changes in ethnic-racial identity compare when teachers versus researchers implement the program.

The Significance of Teachers for Youth ERI Development and Adjustment

Teachers play a critical role in students' lives, including in their identity development (Verhoeven et al., 2019). Furthermore, teachers can positively impact students' psychological and academic adjustment through the provision of various types of social support (e.g., instrumental, emotional, informational; for a review see Sterrett et al., 2011). There is also empirical evidence demonstrating the positive impact that educators' engagement in culturally sustaining pedagogical practices can have on students' academic adjustment (see Aronson & Laughter, 2016, for a review), and conceptual work suggesting that curricula such as the *Identity Project* can provide a useful entry point for teachers to engage in culturally sustaining pedagogical practices with their students (Umaña-Taylor, 2023a). To be clear, the focus on supporting students' ethnic-racial identity exploration in the *Identity Project* is likely a necessary, but not sufficient, curricular approach to bring culturally sustaining pedagogical practices into the classroom.

Culturally sustaining pedagogical practices center students' languages, literacies, and ways of being in all aspects of teaching (e.g., curriculum, instruction, assessment) and view students' ethnoracial backgrounds as strengths that can not only be nurtured and fostered in the classroom, but that can also be leveraged to make the classroom a more powerful context for learning (Alim & Paris, 2017). Building on foundations initially introduced in culturally relevant pedagogy (Ladson-Billings, 1994) and culturally responsive teaching (Gay, 2000), culturally sustaining pedagogy seeks to bring students' ethnoracial identities into the classroom, replace deficit perspectives of students and their ethnoracial communities with strengths-based perspectives, and critically center the strengths, assets, and communities of

² Although the *Identity Project* consists of 8 sessions that are delivered once per week, implementation took place over a 10-week period because of two non-consecutive school holiday weeks that interrupted the weekly implementation. Thus, pretest was administered in Week 1, *Identity Project* sessions were delivered during Weeks 2 through 11, and the posttest was administered in Week 12.

students of color in the classroom (Paris, 2021). Differences between culturally sustaining pedagogy and other conceptualizations, however, are that culturally sustaining pedagogical practices consider sustaining and fostering students' cultural ways of knowing and being a key goal of schooling; furthermore, with culturally sustaining pedagogy there is a strong emphasis on recognizing that students' and communities' cultural ways of being are dynamic and constantly evolving, and that it is essential to support youth in thinking critically about their cultural practices – both in terms of honoring them *and* critically questioning whether their practices reproduce or perpetuate other forms of oppression (e.g., homophobia, sexism, ableism) that may unintentionally marginalize others (Alim et al., 2020). Thus, in addition to valuing students' ethnoracial cultural backgrounds, culturally sustaining pedagogical approaches require students to reflect deeply on their cultural practices, consider them in context, and think critically about ways in which there is value in their practices as well as ways in which transformations may be necessary to avoid perpetuating inequities based on other social identities. Ultimately, culturally sustaining pedagogical approaches frame students as experts in their cultures and expect that, as experts, they also will engage in deep reflection regarding their culturally informed strengths and areas for improvement.

Regarding the impact of culturally sustaining pedagogical practices on student outcomes, one study found that teachers' implementation of an ethnic studies curriculum (which centered culturally sustaining practices such as engaging students in dialogue, critical thinking, and writing about Mexican-American, African American, and Asian American histories; discussing ethnoracial inequality and injustices; providing explicit opportunities for students to connect their identities and lived experiences to the curriculum; and prioritized student-led discussions) led to increases in students' confidence in academic, social, and cultural domains (Nojan, 2020). Similarly, other studies have documented positive associations between educators' engagement in culturally sustaining pedagogical practices (e.g., connecting students' ethnoracial backgrounds to course content, communicating a culture of care and authentic relationships, and structuring assignments and lessons in a manner that engages students in critical thinking and analysis regarding ethnoracial marginalization and the educational landscape) and academic-related outcomes in math and science (Rodriguez et al., 2004), English (Christianakis, 2011), and social studies (Choi, 2013).

The *Identity Project* curriculum can provide an important entry point for teachers to engage in this work with students in their classrooms by promoting students' engagement in the process of ethnoracial exploration with structured activities, providing opportunities for students to recognize their agency and expertise related to their own identity

through activities in which they share what they are learning with one another and with their teachers, and ultimately by fostering relational trust and building community. Indeed, fostering trust and community are essential for students to feel safe and willing to engage in critical discussion (Sleeter & Zavala, 2020). Furthermore, exploration is a necessary foundational step to move toward the exercise of critically evaluating the strengths and potential challenges of students' cultural expressions, and the *Identity Project* can help establish the conditions within which this can happen in a classroom setting. Indeed, a recent study of 9th and 10th grade English and History teachers found that teachers' use of culturally sustaining pedagogical practices (e.g., lessons are contextualized in students' identities and experiences; mutual respect and healthy boundaries are established between the teacher and their students) over the course of an academic year was more prevalent when they were teaching the *Identity Project* curriculum, relative to their standard curriculum (Satterthwaite-Freiman et al., 2024). Given their considerable influence on students, teachers may not only positively impact students' ethnic-racial identity development as they implement the *Identity Project* curriculum, but also the effects of the curriculum could be stronger when delivered by teachers than when delivered by researchers. This represents a significant gap in the literature, as it is currently unknown if teachers' implementation of the *Identity Project* will produce similar increases in students' ethnic-racial identity exploration as those achieved when researchers deliver the intervention.

Testing Variability by Family Ethnic-Racial Socialization, Ethnoracial Background, and Student-Teacher Ethnoracial Majority/Minoritized Status Match

In addition to evidence from experimental work demonstrating that students' ethnic-racial identity exploration can be promoted through the *Identity Project* intervention (Umaña-Taylor, Douglass et al., 2018), family ethnic-racial socialization is another powerful influential factor for students' ethnic-racial identity development. A plethora of studies have demonstrated that families' explicit and implicit messages to youth regarding their ethnoracial background are significantly and positively associated with youth engaging in greater exploration and developing a greater sense of clarity regarding their ethnic-racial identities (see Huguley et al., 2019, for a meta-analysis). Indeed, in a follow-up analysis to the original efficacy trial of the *Identity Project*, family ethnic-racial socialization was found to enhance intervention effects on ethnic-racial identity exploration, such that the positive effect of the intervention on ethnic-racial identity exploration was almost five times greater for youth reporting higher levels ($d = 0.887$) versus lower levels ($d = 0.178$) of family ethnic-

racial socialization (Sladek et al., 2021). As such, the current study hypothesized that increases in ethnic-racial identity exploration would be significantly stronger for youth who reported higher levels of family ethnic-racial socialization at the start of the intervention, relative to those who reported lower levels of family ethnic-racial socialization.

In addition to examining family ethnic-racial socialization as a potential moderator of change, an exploratory research question to examine whether ethnoracial majority vs. minoritized background would moderate change in ethnic-racial identity exploration was pre-registered. The *Identity Project* was designed as a universal health promotion program, and though the theoretical and empirical literature supports mean level differences in ethnic-racial identity developmental processes between youth from ethnoracial majority versus minoritized backgrounds due to ethnicity-race being more salient to members of minoritized groups, the impact of the *Identity Project* for prompting increases in exploration has been found to be comparable across ethnoracial majority and minoritized youth (Umaña-Taylor, Douglass et al., 2018). As such, this exploratory research question was included in the study design to continue to build evidence regarding whether experiences with the *Identity Project* varied for students from ethnoracial majority versus minoritized backgrounds.

Finally, although students' ethnoracial background was not expected to differentially inform the effects of teachers' implementation of the *Identity Project* on students' ethnic-racial identity exploration, a different possibility was that moderation could emerge based on the match or mismatch between students' and teachers' ethnoracial backgrounds. Prior research has found that the quality of relational interactions between students and teachers are more positive and less negative when students and teachers are ethnoracially matched (Battey et al., 2018). Furthermore, students taught by a demographically similar teacher (i.e., ethnoracial background and gender) report greater personal effort in school, higher feelings of care from their teachers, better quality student-teacher communication, higher happiness in class, and a greater sense of being motivated by their teachers (Egalite & Kisida, 2018). Although much of this research has focused on students and teachers who are racially Black or White, there is some evidence that students of color benefit from being matched with a teacher of color, even if they do not share the same ethnoracial background (e.g., Yarnell & Bohmstedt, 2018).

Related to the *Identity Project*, specifically, students of color who are taught by teachers of color, relative to those taught by White teachers, may be less hesitant to engage with a curriculum focused on ethnicity-race; students of color may also potentially perceive more support or understanding from teachers of color, relative to White teachers, when sharing their experiences related to an ethnoracial minoritized identity, given they might see a teacher from an ethnoracial

minoritized background as someone who could relate to their experiences. Together, these factors could ultimately lead to more authentic engagement for students of color when paired with a teacher of color. For White students, being taught the *Identity Project* by White teachers could be particularly impactful because they are witnessing a White person who is in a position of power and authority in their school communicating that it is relevant and important as a White-identifying person to examine ethnicity-race as a part of one's identity. In contrast to a teacher of color leading this activity, White students may identify more with a White teacher due to their shared ethnoracial majority status and therefore be more willing to explore this part of their identity or believe that it is a worthwhile exercise. As such, it is possible that an ethnoracial majority/minoritized student-teacher match may result in students being more motivated to engage with the *Identity Project* curriculum and perhaps show greater changes in ethnic-racial identity exploration relative to those experiencing a student-teacher mismatch.

An equally viable possibility, however, is that being taught the *Identity Project* curriculum by a teacher who *does not* share one's ethnoracial majority/minoritized status may be just as powerful as being taught by someone who does share this demographic characteristic – the reasons for this, however, are slightly different for students of color and White students. For students of color, whose experiences in the education system have been historically and in contemporary times characterized by marginalization, having a White teacher (who is a member of the ethnoracial majority group and is in a position of power and authority in the school setting) center ethnic-racial identity as part of the regular school curriculum can be a powerful experience. For example, by engaging students in discussions and activities that acknowledge the inaccuracies and consequences of ethnoracial stereotypes, and leading students through a curriculum that empowers them to explore and define issues of ethnicity-race on their own terms, a White educator is communicating to students from ethnoracial minoritized backgrounds that their backgrounds and experiences matter. This may provide some students of color with their first experience of a White educator truly 'seeing' them and sending the message that all parts of who they are (including their ethnic-racial identity) are important. For White students, who oftentimes do not realize that they have an ethnic-racial identity and when faced with experiences that prompt them to consider their White ethnic-racial identity can be left with feelings of confusion, guilt, or lack of understanding (Moffitt & Rogers, 2022), this can be a powerful message coming from an educator of color who is in a position of power and authority in their school confirming that they, too, have an ethnic-racial identity that is important to explore and understand. Given the competing nature of these possibilities, a non-directional hypothesis

was pre-registered to examine if and how students' changes in ethnic-racial identity exploration varied as a function of student-teacher ethnoracial match or mismatch.

The Current Study

Given the lack of empirical evidence on teachers' implementation of ethnic-racial identity curricula, the current study examined changes in adolescents' ethnic-racial identity exploration as a function of their 9th grade U.S. History teachers' implementation of the *Identity Project*. The first pre-registered hypothesis (see <https://doi.org/10.17605/OSF.IO/W6NPC>) was that ethnic-racial identity exploration would increase from pretest to posttest. Second, it was hypothesized that increases in ethnic-racial identity exploration would be more pronounced for adolescents with higher levels of family ethnic-racial socialization at baseline, relative to adolescents with lower levels of family ethnic-racial socialization. One non-directional moderation hypothesis was pre-registered, which posited that ethnic-racial identity exploration from pretest to posttest would vary as a function of student-teacher ethnoracial match. Finally, the pre-registration included an exploratory moderation analysis focused on testing whether changes in ethnic-racial identity exploration from pretest to posttest would vary by students' ethnic-racial majority or minoritized status background.

Method

Participants

Data were from a longitudinal study of 9th grade adolescents ($N = 180$; $Mage = 14.11$ years; $SD = 0.33$; range 13–15; 86.1% U.S.-born; 71.1% at least one foreign-born parent) attending a high school in the U.S. Northeast in which their 9th grade U.S. History teachers were implementing the *Identity Project* curriculum during the 2021–2022 academic year. Consistent with the research question, data gathered in Sept–Oct 2021 (T1), prior to the first *Identity Project* session (i.e., pretest), and Dec 2021 (T2), after completion of all *Identity Project* sessions (i.e., posttest) were examined. In terms of retention, 82.2% of participants ($n = 148$) completed T2 surveys.

With respect to primary ethnoracial background, participants self-identified as Latinx³ (38.3%), White (33.9%),

Black (15.0%), Asian American (9.4%) and another ethnoracial background (e.g., Arab, Native American; 3.3%). The ethnic-racial composition of the sample closely mirrored the composition of the school population. Furthermore, participants self-identified as female (50.6%), male (42.2%), and another gender identity (e.g., non-binary, gender-fluid; 6.7%); 0.6% did not respond. Regarding parents' educational attainment, participants indicated that their mothers had less than a high school degree (21.1%); at least a high school education or equivalent (16.7%); some college education (12.8%); a degree from a 2 or a 4-year institution (16.7%); or a master's or professional degree (18.3%). Some participants did not know (16.7%) or did not report their mothers' educational attainment (1.7%). Participants reported that their fathers had less than a high school degree (15.0%); at least a high school education or equivalent (12.8%); some college education (10.6%); a degree from a 2 or a 4-year institution (21.7%); or a master's or professional degree (12.2%). Some participants did not know (26.1%) or did not report their fathers' educational attainment (1.7%).

Procedure

All 9th grade students enrolled in U.S. History experienced the *Identity Project*, which was implemented by their U.S. History teachers approximately once per week. To prepare for the *Identity Project*, teachers participated in 32 h of professional development, which included participating in the *Identity Project* sessions that they would eventually implement with their students. The professional development was guided by four learning domains that were interwoven throughout the program: ethnic-racial identity content knowledge, personal ethnic-racial identity development, ethnoracial-based systemic inequities and the U.S. education system, and facilitation strategies for engaging in discussions about race, ethnicity, and identity with students. The training included continuous personal reflection, many opportunities for sharing with colleagues, minimal didactic instruction, and many activities (e.g., pair, small group, large group) to encourage and facilitate active learning.

Participation in survey data collection was voluntary and all 9th grade students in the teachers' U.S. History courses were eligible to participate. A total of 16 classes, distributed across 4 teachers, experienced the *Identity Project*. Teachers provided information letters and consent forms for students to bring home to their parents/guardians; all materials were made available to parents/guardians in English, Spanish, Portuguese, Haitian Creole, and Mandarin. Students who returned the signed forms, regardless of their decision to participate, received a \$10 gift card. The research team sent unique T1 survey links via email and text message to all

³ The response option was "Latino or Hispanic (for example: Mexican, Guatemalan, Salvadoran, Colombian, etc.);" however, the term Latinx is used in the current article to promote terminology that recognizes gender as non-binary.

students with parental consent forms. Students were asked to complete assent forms electronically on the first page of the survey. Parental consent was obtained from 186 (62%) of all eligible students, and 180 completed the T1 survey. Survey data collection took place online, outside of school hours, and lasted approximately 30 min per survey. After teachers completed the last *Identity Project* session, students who had participated at T1 received unique T2 survey links via email and text message. All participants completed surveys in English and received a \$15 gift card for each survey in which they participated. All procedures were approved by the Harvard University institutional review board and the school district.

Measures

Ethnic-racial identity exploration

Adolescents' ethnic-racial identity exploration was assessed at T1 and T2 with the exploration subscale (3 items; e.g., "I have attended events that have helped me learn more about my ethnicity-race") of the Ethnic Identity Scale-Brief (Douglass & Umaña-Taylor, 2015). Previous work has provided support for the validity and reliability of this subscale with an ethnoracially diverse sample of adolescents (Umaña-Taylor, Douglass et al., 2018). Response options ranged from 1 (does not describe me at all) to 4 (describes me very well). Means were computed with higher scores reflecting higher ethnic-racial identity exploration. Cronbach's α were 0.70 at T1 and 0.78 at T2.

Ethnoracial background

At T1, adolescents were asked to choose the category or categories that best described their ethnoracial background using the following list: Black/African American, Latino or Hispanic, Asian, American Indian/Native American, White, and other). Adolescents who chose more than one option were asked a follow-up question asking them to identify the ethnic-racial category that they felt most a part of, using the same list as above. For analyses, using adolescents' response to the first question (or based on their response to the follow-up question, if they selected multiple options for the first), ethnoracial background was coded with students of color as the reference group: 0 = Black, Latinx, Asian, American Indian, or Other, $n = 119$ or 1 = White ($n = 61$).

Family ethnic-racial socialization

At T1, adolescents reported on the degree to which they perceived that their families had socialized them with respect to their ethnic-racial culture in the past 30 days with

a 7-item (e.g., "My family teaches me about my ethnic-racial background;" response options ranged from 1 = not at all to 5 = very much) shortened version of the Family Ethnic Socialization Measure (Umaña-Taylor et al., 2004). Items were selected based on psychometric analyses with data collected from a different sample of ethnoracially diverse adolescents exploring indicators of reliability (e.g., alphas, inter-item correlations), validity (e.g., correlations with theoretically related constructs such as ethnic-racial identity, discussion among researchers regarding items with maximum face validity), and scale means and variability. For example, items with relatively lower inter-item correlations were removed if the alpha value remained high and if removing these items did not drastically affect the mean and variability of scale scores. These psychometric indices were comparable in the original 12-item scale and the 7-item shortened version overall, and also specifically when examined separately for Asian American, Black, Latinx, and White adolescents. Previous work has provided evidence for the validity and reliability of the full scale with an ethnoracially diverse sample of adolescents (Sladek et al., 2020). With the current sample, a one-factor confirmatory factor analysis model demonstrated good fit with this shortened 7-item version: $\chi^2(14) = 35.02$, $p = 0.002$, CFI = 0.96, RMSEA = 0.09 [90% CI: 0.06, 0.13], SRMR = 0.04; all standardized factor loadings > 0.65. Means were computed and higher scores reflected higher family ethnic-racial socialization. Cronbach's α was 0.90 at T1.

Student-teacher ethnoracial match

Student-teacher ethnoracial match was coded using teachers' ethnoracial self-identification and students' ethnoracial background, described above. For the purposes of analyses, student-teacher ethnoracial match was coded as 0 = match ($n = 121$), 1 = mismatch ($n = 59$). Students of color being taught exclusively by a teacher of color were considered a match ($n = 51$), and White students being taught by a White teacher were also considered a match ($n = 35$). Students coded as mismatch included students of color being taught exclusively by a White teacher ($n = 38$) and White students being taught exclusively by a teacher of color ($n = 21$). In cases where students were in a class that was co-taught by a teacher of color and a White teacher, they were coded as a match ($n = 5$ White students, $n = 30$ students of color).

Additional sociodemographic variables

For analyses, gender was coded 0 = female or other gender ($n = 103$); 1 = male ($n = 76$). Immigrant generational status was coded as a sum of family members born in the U.S., including the participant, the participant's

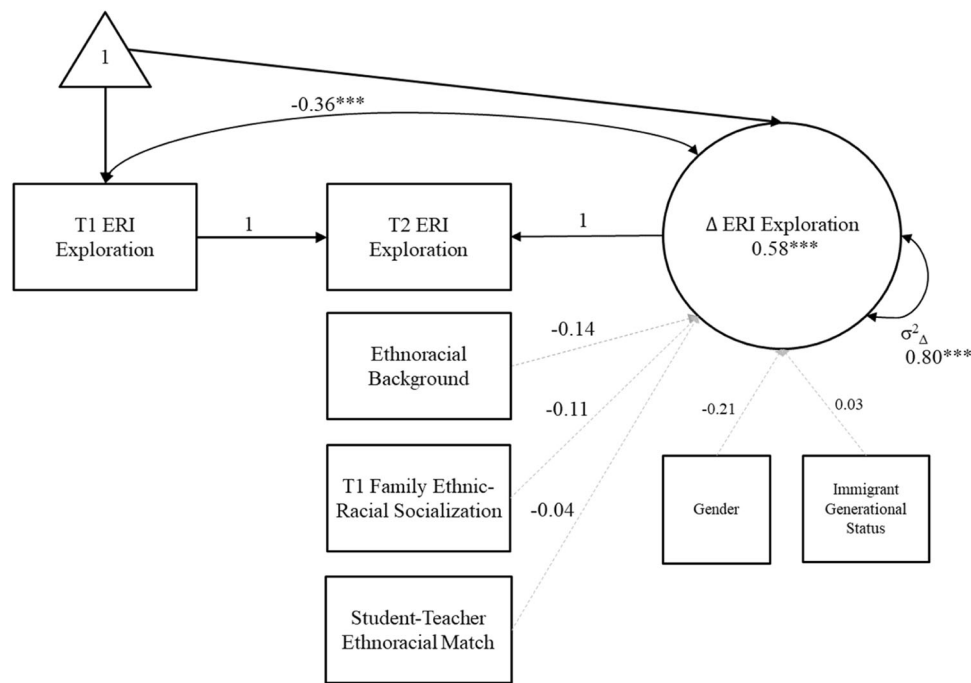


Fig. 1 Univariate two-occasion latent difference score model of ethnic-racial identity exploration changes. *Note.* $N = 180$. ERI = ethnic-racial identity. Unstandardized estimates. Variances and covariances among T1 exogenous variables were estimated and omitted from figure for ease of presentation. Gray dashed lines indicate non-significant paths ($p > 0.05$). Ethnoracial background was coded: 0 = Latinx, Black, Asian American, or another ethnoracial minoritized background; 1 =

White. Student-Teacher ethnoracial match was coded: 0 = match; 1 = mismatch. Gender was coded 0 = female or another gender identity label; 1 = male. Immigrant generational status was coded as a sum of family members born in the U.S., including the participant, the participant's parents, and both sets of grandparents (0 = none born in the U.S. to 7 = all born in the U.S.). *** $p < 0.001$

parents, and both sets of grandparents (0 = none born in the U.S. to 7 = all born in the U.S.).

Analytic Strategy

Missing data ranged from 0.0% to 26.1% across study variables, and results from Little's MCAR test using SPSS 29 suggested that data were missing completely at random, $\chi^2(16) = 23.56$, $p > 0.05$. Hypotheses were tested in *Mplus* v8 (Muthén & Muthén, 2017) using a latent difference score (LDS) model (McArdle & Hamagami, 2001) within a structural equation modeling framework with full information maximum likelihood (FIML) estimation to address missing data. Because intervention effects could be modified by sociodemographic factors (i.e., student gender, immigrant generational status; Umaña-Taylor & Douglass, 2017), each of these were included as covariates in the analyses. The LDS approach was selected due to its strengths for using latent variable modeling to obtain reliable estimates of intra-individual change across two occasions and addressing measurement error issues that are introduced when relying on observed difference scores to test change between two repeated measures (see Castro-Schilo & Grimm, 2018). Fitting an LDS model allowed for

simultaneously estimating the latent change in ethnic-racial identity exploration from T1 to T2 (Δ ethnic-racial identity exploration), between-person variance in this change (σ^2_{Δ}), and moderators (i.e., T1 family ethnic-racial socialization, student ethnoracial background, student-teacher ethnoracial match) and covariates (i.e., student gender, immigrant generational status) of changes in ethnic-racial identity exploration, while accounting for concurrent associations among T1 variables (e.g., ethnic-racial identity exploration and family ethnic-racial socialization; see Fig. 1). The continuous variables T1 family ethnic-racial socialization and immigrant generational status were grand mean centered. Univariate two-occasion LDS models are saturated, and thus, conventional SEM fit indices (e.g., CFI, RMSEA) are not applicable. Significance testing focused on whether parameters of interest (e.g., Δ ethnic-racial identity exploration) were significantly different from zero. Intraclass correlations (ICCs) indicated less than 1% of the variance in ethnic-racial identity exploration at T1 and T2 were attributable to classroom-level or teacher-level differences; thus, multi-level modeling to partition cluster-level variance was not suitable for these analyses. Finally, effect size for mean level changes in ERI exploration from T1 to T2 was

Table 2 Bivariate correlations and descriptive statistics for focal variables

	1	2	3	4	5	6	7
1. T1 ERI Exploration	--						
2. T2 ERI Exploration	0.31***	--					
3. Ethnoracial background	−0.24**	−0.20*	--				
4. T1 Family ethnic-racial socialization	0.53***	0.38***	−0.43***	--			
5. Student-Teacher ethnoracial match	0.02	0.01	0.03	−0.08	--		
6. Gender	0.04	−0.07	−0.04	0.08	−0.02	--	
7. Immigrant generational status	−0.24**	−0.12	0.70***	−0.45***	0.04	0.03	--
M^a (SD)	2.13 (0.77)	2.58 (0.77)	0.34	3.20 (1.00)	0.33	0.42	2.80 (2.63)
n	176	144	180	169	180	179	133

$N = 180$. ERI Ethnic-racial identity. Ethnoracial background was coded: 0 = Latinx, Black, Asian American, or another ethnoracially minoritized background; 1 = White. Student-Teacher ethnoracial match was coded: 0 = match; 1 = mismatch. Gender was coded 0 = female or another gender identity label; 1 = male. Immigrant generational status was coded as a sum of family members born in the U.S.A., including the participant, the participant's parents, and both sets of grandparents (0 = none born in the U.S.A. to 7 = all born in the U.S.A.)

^aPercentage reported for dichotomous variables

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

calculated using Hedge's g (equivalent to Cohen's d for samples greater than 20; Morris & DeShon, 2002) using a publicly available online calculator for effect size with paired-samples t -test (for access to the tool and formulas, see <https://effect-size-calculator.herokuapp.com/>).

Results

Preliminary Analyses and Descriptive Statistics

Study attrition was not significantly associated with participants' ethnoracial background, $\chi^2(1) = 0.12$, $p = 0.73$, T1 ethnic-racial identity exploration, $t(56.15) = 1.27$, $p = 0.21$, T1 family ethnic-racial socialization, $t(167) = 1.44$, $p = 0.15$, student-teacher ethnoracial match, $\chi^2(1) = 1.07$, $p = 0.30$, gender, $\chi^2(1) = 0.03$, $p = 0.87$, or immigrant generational status, $t(131) = -0.85$, $p = 0.40$. Bivariate correlations and descriptive statistics are presented in Table 2. Consistent with prior research, students of color reported significantly higher ethnic-racial identity exploration compared to White students at T1, $r(174) = -0.24$, $p < 0.01$, and T2, $r(142) = -0.20$, $p = 0.02$. T1 family ethnic-racial socialization was positively correlated with ethnic-racial identity exploration at T1, $r(166) = 0.53$, $p < 0.001$, and T2, $r(141) = 0.38$, $p < 0.001$. Student-Teacher ethnoracial match was not significantly correlated with ethnic-racial identity exploration at T1, $r(174) = 0.02$, $p = 0.84$, or T2, $r(142) = 0.01$, $p = 0.87$. In this sample, students of color and White students were equally likely to have a teacher of color or a White teacher, $\chi^2(1) = 0.11$, $p = 0.74$.

Testing Change in Ethnic-Racial Identity Exploration

LDS model results are reported in Fig. 1. Ethnic-racial identity exploration significantly increased from T1 to T2, $\Delta = 0.58$ ($SE = 0.14$), $p < 0.001$, Cohen's $d = 0.52$ (medium effect size), and there was significant variability in ethnic-racial identity exploration changes, $\sigma^2_{\Delta} = 0.80$ ($SE = 0.09$), $p < 0.001$ (see Fig. 2). Furthermore, students who started out with lower T1 ethnic-racial identity exploration showed a significantly steeper increase in ethnic-racial identity exploration over time, $\text{cov}(\text{intercept}, \Delta) = -0.36$ ($SE = 0.05$), $p < 0.001$. Changes in ethnic-racial identity exploration did not vary based on student ethnoracial background, $b = -0.14$ ($SE = 0.23$), $p = 0.54$, T1 family ethnic-racial socialization, $b = -0.11$ ($SE = 0.08$), $p = 0.20$, student-teacher ethnoracial match, $b = -0.04$ ($SE = 0.15$), $p = 0.81$, gender, $b = -0.21$ ($SE = 0.15$), $p = 0.16$, or immigrant generational status, $b = 0.03$ ($SE = 0.04$), $p = 0.47$. Collectively, these variables did not account for significant variance in ethnic-racial identity exploration changes, $R^2 = 0.03$ ($SE = 0.03$), $p = 0.24$.

Sensitivity Analyses

Multigroup modeling was utilized to examine whether the extent to which (a) T1 family ethnic-racial socialization predicted changes in ethnic-racial identity exploration varied by student ethnoracial background (i.e., differences for students of color and White students) and (b) changes in ethnic-racial identity exploration varied by the type of student-teacher ethnoracial match (i.e., differences for students of color with a teacher of color, $n = 81$, students of color with a White

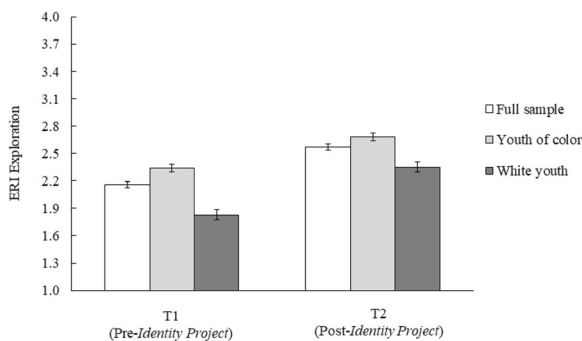


Fig. 2 Ethnic-racial identity exploration changes from before to after the Identity Project. *Note.* Means are from paired-sample *t*-tests. All differences between T1 and T2 were statistically significant, $p < 0.01$. Error bars are standard error of the mean. Youth of color were Latinx, Black, Asian American, or from another ethnoracially minoritized background

teacher, $n = 38$, White students with a teacher of color, $n = 21$, and White student with a White teacher, $n = 40$). Using student ethnoracial background as the grouping variable, comparing a freely estimated model with a model constraining all parameters to be equal for students of color and White students indicated there were no significant differences, $\chi^2(9) = 6.86$, $p = 0.65^4$. Furthermore, comparing a freely estimated model with a model constraining the path from T1 family ethnic-racial socialization to the latent change in ethnic-racial identity exploration to be equal indicated there was not a significant difference between students of color and White students, $\chi^2(1) = 0.04$, $p = 0.85$. Using the four types of student-teacher ethnoracial match as the grouping variable, comparing a freely estimated model with a model constraining all parameters to be equal across these groups indicated there were no significant differences, $\chi^2(21) = 13.06$, $p = 0.91$. Furthermore, comparing a freely estimated model with a model constraining the latent intercept of ethnic-racial identity exploration change (Δ) to be equal across these groups indicated there were no significant differences, $\chi^2(3) = 1.98$, $p = 0.58$. Thus, the changes in ethnic-racial identity exploration from before to after the *Identity Project* did not vary with combinations of ethnoracial match between students and their teachers.

Discussion

Adolescents' ethnic-racial identity development is an important developmental competency that is positively associated with adjustment and negatively associated with maladjustment. Ethnic-racial identity becomes particularly

salient during adolescence, and it can play an important role in reducing ethnoracial disparities through various mechanisms, such as by protecting ethnoracial minoritized youth from the negative effects of ethnoracial-based threats and promoting more positive intergroup contact attitudes and relationships (Umaña-Taylor, 2023b). The current study provides promising preliminary evidence that it is possible to train educators to increase adolescents' engagement in the ethnic-racial identity developmental process through a school-based curriculum. Although based on non-experimental data, the current findings suggest that teachers may be efficacious facilitators of the *Identity Project* curriculum. Furthermore, when considered in relation to prior researcher-led implementations, changes in students' ethnic-racial identity exploration from before to after experiencing the *Identity Project* curriculum appear to be considerably larger when teachers deliver the program (i.e., Cohen's $d = 0.52$) compared to researchers (Cohen's $d = 0.13$, Ceccon et al., 2023; Cohen's $d = 0.17$, Umaña-Taylor, Douglass et al., 2018)⁵, underscoring the potential power of educators for promoting students' identity development.

Indeed, the significant impact of educators as deliverers of the curriculum may help explain why the second hypothesis (i.e., changes in exploration were expected to be stronger for adolescents with higher, relative to lower, levels of family ethnic-racial socialization) was not supported in the current study. In the original efficacy trial, which was delivered by researchers, the intervention effect was significantly stronger among youth who reported higher levels of family ethnic-racial socialization, suggesting that those who might benefit most from the intervention or be more likely to engage with the school-based curriculum were those who had some experiences with ethnic-racial socialization in the family context (Sladek et al., 2021). Perhaps this extrinsic motivator for students' engagement with the curriculum (i.e., a congruence between the home and school context) is necessary when researchers, who are relatively unknown to students and do not have an existing relationship with them when the curriculum begins, are implementing the curriculum. In contrast, the closeness of the student-teacher relationship, the respect for teachers as authority figures in the school context, the skill with which

⁴ The mean of T1 family ethnic socialization was set to zero, given that the variable was centered and thus the mean was already equal to zero. Applying this constraint to the fully saturated model allowed for 1 *df* for these nested model comparisons.

⁵ To ensure accurate effect size comparisons for researcher-led implementations, which were based on experimental designs and therefore included comparison to a control group, Hedge's g (equivalent to Cohen's d for samples greater than 20) was calculated for the two prior published studies and the current study with pre- and post-test means only considering youth in the intervention group (i.e., repeated measures); see Morris and DeShon (2002) for a review of effect size equivalence considerations. Thus, the effect sizes reported here differ slightly from effect sizes reported in Ceccon et al. (2023) and Umaña-Taylor, Douglass et al. (2018), which were based on experimental results comparing intervention and control group differences.

teachers deliver lessons – or a combination of some or all of these – may help explain why family ethnic-racial socialization did not materialize as an important source of variability in the changes that emerged in ethnic-racial identity exploration in the current study. Future studies with larger samples and using an experimental design will be important to more rigorously test these ideas. For example, in considering the prior experimental results (Sladek et al., 2021) and those of the current pre/posttest analysis, evidence for the notion that family ethnic-racial socialization promotes adolescents' ethnic-racial identity exploration in the school-based *Identity Project* may specifically emerge when comparing youth who complete the program to those in a control group (who have little to no introduction of these concepts during school time) – even when teachers are implementing the curriculum.

Why Train Educators?

The *Identity Project* was developed with the intention of dissemination in school-based settings, given the compulsory nature of schooling and the desire to avoid selection bias whereby only youth willing to engage with the sensitive topic of race would participate; furthermore, school settings provide an ideal context for scale-up by enabling broader reach to the intended population (Umaña-Taylor & Douglass, 2017). However, beyond these important structural factors that make implementation of the *Identity Project* in school-based settings ideal, engaging students' teachers as facilitators of the curriculum has the potential to significantly enhance the effects of the curriculum – which is what preliminary findings from the current study, when compared to prior researcher-based implementations, suggest.

It is well-documented that educators play a significant role in the lives of their students and can positively influence their academic and psychological outcomes in myriad ways (Sterrett et al., 2011). However, beyond teachers' influence due to structural features such as being in a referential position of authority and spending a considerable amount of time with youth, educators communicate important ethnoracial-based messages (often unconsciously; Benson & Fiarman, 2021) to students from all ethnoracial backgrounds regarding whether they belong in the school (Umaña-Taylor, 2023b). These messages and unconscious biases can have a significant impact on students' academic adjustment, and a particularly negative impact on ethnoracial minoritized youth's academic adjustment (Okonofua et al., 2016). In contrast, when educators take a culturally sustaining approach in their work with students, students' identities are centered within the academic curriculum in a manner that communicates their value and worth; and such practices have been demonstrated to positively impact student outcomes

(e.g., Bonilla et al., 2021). It is possible that the *Identity Project* curriculum can facilitate teachers' engagement in culturally sustaining teaching practices by providing them with a manualized curriculum consisting of activities that enable explicitly embedding students' identities into the classroom, communicating to students that the funds of knowledge they bring to the school setting are valuable and an important part of the lesson, and inviting students to see themselves as experts and agents in informing the curriculum.

Training educators to implement the *Identity Project* can not only facilitate educators' ability to engage in culturally sustaining approaches in their classroom but could also have positive effects on students' adjustment by reducing unintentional, yet impactful, student experiences with ethnoracial marginalization. For instance, the *Identity Project* teacher training engages educators in activities that increase their awareness of ethnoracial-based systemic inequities and their downstream impacts on students' academic and psychological adjustment and provides educators with opportunities to reflect on existing strategies and learn new strategies for facilitating classroom conversations about race, ethnicity, and identity – all with an explicit focus on fostering a climate of ethnoracial inclusion and equity (Umaña-Taylor, 2023b). Thus, in addition to the benefits that can come from students' personal experiences with the *Identity Project*, there may be additional mechanisms by which students' positive outcomes are augmented when teachers are involved in the implementation. This is an important question to explore in future research.

In sum, the current findings, which provide preliminary evidence that students' ethnic-racial identity exploration increased significantly over the course of approximately 12 weeks during the time that their teachers were implementing the *Identity Project* in their classrooms, add further support to the notion that teachers can have a significant impact on students' lives (Aronson & Laughter, 2016; Sterrett et al., 2011). However, there are several limitations that limit the conclusions that can be drawn from the current findings, and further work is necessary with more rigorous research designs to draw more definite conclusions.

Limitations and Directions for Future Research

Findings from the current study provided important insights into whether ethnic-racial identity exploration would change after a teacher-led implementation of the *Identity Project*; however, there are limitations to acknowledge that provide important directions for future research. First, the current study did not use an experimental design, such as a randomized controlled trial; thus, the findings are limited to informing an understanding of prospective longitudinal changes in ethnic-racial identity exploration in a sample of adolescents whose teachers were implementing the *Identity*

Project curriculum. Without the ability to understand these findings in relation to a control group, the possibility remains that the significant increases that were evidenced in students' ethnic-racial identity exploration in the current study were the result not of teachers' implementation of the *Identity Project*, but of unobserved variables (e.g., community events increasing students' awareness of ethnoracial diversity) that coincided with the timing of teachers' implementation of the *Identity Project*. Given this limitation, the current findings cannot be used to imply causality, should not be considered conclusive, and instead should be interpreted as preliminary and suggestive of the potential for teachers' implementation to affect students' ethnic-racial identity exploration. It will be important to test the questions examined in the current study with a design that includes both an intervention and a control group. Furthermore, the current study focused exclusively on change in the immediate intervention target of the *Identity Project* (i.e., exploration) at the 12-week posttest, which is consistent with the theory of intervention but provides a limited understanding of the longer-term cascading effects of the curriculum on ethnic-racial identity resolution, global identity cohesion, and indices of adjustment. Thus, future studies using an experimental design should also examine the extent to which the *Identity Project*, when delivered by teachers, has long-term and downstream impacts on adolescents' ethnic-racial identity and adjustment.

Another noteworthy limitation is that teachers' fidelity of implementation of the *Identity Project* was not monitored; thus, the extent to which teachers delivered the program with fidelity is unknown and this could introduce important variability into the current results. Put differently, the effects reported here could be a conservative estimate of change in students' exploration, as effect sizes could be larger among students whose teachers implemented with greater fidelity. This will be an important possibility to explore in future research.

Another important limitation pertains to the limited generalizability of the current findings due to the reliance on a convenience sample representing students and teachers from one school site located in a northeastern U.S. state that is characterized by relatively more progressive attitudes regarding ethnoracial issues, as evidenced by the absence of state legislation that explicitly prohibits conversations about race, ethnicity, and identity in K12 or higher education settings. Findings could vary significantly in other geographical regions, such that changes in students' ethnic-racial identity exploration could be even larger in contexts where there is greater censorship regarding these topics, as the increased exposure in school could lead to heightened awareness and salience of ethnic-racial identity and prompt even greater exploration. Furthermore, in contexts where such conversations are explicitly prohibited, educators who are willing to engage in such discussions may be particularly invested in

supporting youth as they think about issues of ethnic-racial identity, which could also result in stronger program effects. These are important sources of variability to explore in future studies, especially given the vast diversity in sociopolitical attitudes related to addressing issues of race and ethnicity in schools currently evident across the U.S. (PEN America, 2022) and in other countries (e.g., Schachner et al., 2023).

Finally, given the relatively small sample size, there was limited variability in important sociodemographic factors that could be important moderators of program effects. For instance, most students in the current study were U.S. born. Although a significant proportion of the sample represented students who had at least one immigrant parent, it will be important to understand whether students' experiences with the *Identity Project* curriculum vary when they are more recent immigrants to the U.S. Adolescents who have arrived in the U.S. within the past couple of years, for example, may be grappling with issues specific to their relatively more recent migration such as becoming aware of and trying to comprehend U.S. notions of race and ethnicity, given these are context-dependent social constructs. The current study did not have the sample size to enable exploration of this variability. Similarly, although the potential of student-teacher ethnoracial match/mismatch to help explain variability in adolescents' ethnic-racial identity exploration changes was tested in the current study, the small sample size for the different student-teacher combinations in these multiple group models may have limited the ability to detect significant effects; having a larger sample size could provide more confidence in the results. Future research is needed with larger samples to continue testing these important subgroup differences and it is important to note that the results from these exploratory analyses are far from conclusive.

Challenges to Future Research due to a Highly Contentious Sociopolitical Climate

Future research, as described above, will require large and diverse samples of students, educators, and school partners. For instance, samples need to be large enough to enable statistical analyses that are adequately powered to test how program effects vary according to various sociodemographic factors and their intersections (e.g., ethnoracial background, immigrant generational status, nativity status, student-teacher ethnoracial match). The current study found no significant variability in changes in exploration as a function of these sociodemographic factors; however, the current sample size was small and analyses were statistically underpowered, particularly for some analyses (e.g., student-teacher ethnoracial match). Furthermore, school settings are complex and characterized by different aspects of diversity (e.g., ethnoracial composition of the student body; rural vs. urban vs. suburban; private/independent vs. public; high vs. low

resource; located in a politically progressive vs. conservative municipality and state); this variability must be empirically examined to understand the contexts within which the *Identity Project* is more or less efficacious. The most significant challenge researchers face in moving this work forward, which requires obtaining the large and diverse samples (of students, teachers, and schools) needed to answer these questions, is the highly contentious sociopolitical environment that impacts educators' and administrators' ability to engage students in discussions about race or ethnicity in the classroom (Umaña-Taylor, 2023b). In the U.S., the practical and legal implications of this sociopolitical resistance are that more than 240 legislative bills have been recently introduced across all regions of the U.S. to explicitly prohibit K12 educators from discussing issues related to race, ethnicity, and identity in the classroom; furthermore, 29 bills that criminalize these conversations have been signed into law across 17 states (PEN America, 2022). As a result, many educators are hesitant to engage due to fear of legal repercussions (Waxman, 2022) or concerns regarding how parents might react (Delale-O'Connor and Graham, 2018), as examples. Scholars have offered recommendations for building strong research-practice partnerships with schools in an effort to support this work (e.g., Villavicencio et al., 2023); however, even in the context of successful partnerships, threats to external validity will exist for all research studies conducted during this current sociohistorical political moment due to inherent selection bias, given that teachers in some states are facing legal barriers to participation and others are not.

Conclusion

There is increasing empirical evidence that supporting adolescents' ethnic-racial identity development in school can promote academic and psychological adjustment, yet there is a significant lack of research regarding whether educators' implementation of ethnic-racial identity curricula can positively impact students' ethnic-racial identity development. The current study provides promising preliminary evidence that it may be possible to train educators to intervene in adolescents' ethnic-racial identity development with a program such as the *Identity Project*, which has been demonstrated to have important positive impacts on adolescents' development and adjustment. Training educators will be an important means to scale up the intervention to reach more students. Regarding next steps, large-scale efficacy trials of the *Identity Project* and the teacher professional development, together, will be necessary to explore not only changes in adolescents' ethnic-racial identity development and how these changes can promote positive psychosocial functioning, but also the extent to which engaging in this work changes teachers' ethnic-racial identity and their use of culturally sustaining pedagogical

practices. The current findings suggest that training educators and supporting them as they engage in this work with their students could have exponentially positive effects on adolescents' ethnic-racial identity development and, in doing so, identify another key target for intervention that can support positive adolescent development and adjustment.

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Authors' Contributions AJU-T conceived the research questions and drafted the manuscript; MRS contributed to conceptualization of the research questions, conducted the statistical analyses, and helped to draft the manuscript; MDS contributed to conceptualization of the research questions and helped to draft the manuscript. All authors collaborated to secure funding, conceive the larger study, manage data collection, and read and approve the final manuscript.

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Data Sharing and Declaration The data analyzed during the current study are not publicly available but are available from the corresponding author on reasonable request. The study hypotheses and analyses were pre-registered via the Open Science Framework (<https://doi.org/10.17605/OSF.IO/W6NPC>).

Compliance with Ethical Standards

Conflict of Interest The authors declare no competing interests.

Ethical Approval This study was approved by the Institutional Review Board of Harvard University and the participating school district. The research was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki and its later amendments.

Informed Consent All study participants provided active parental consent and youth assent for participation.

References

- Alim, S., & Paris, D. (2017). What is culturally sustaining pedagogy and why does it matter? In Paris, D., & Alim, H. S. (Eds.), *Culturally sustaining pedagogies: Teaching and learning for justice in a changing world* (pp. 1–24). Teachers College Press.
- Alim, H. S., Paris, D., & Wong, C. P. (2020). Culturally sustaining pedagogies: A critical framework for centering communities. In

- Suad Nasir, N., Lee, C. D., Pea, R., & McKinney de Royston, M. (Eds.). *Handbook of the cultural foundations of learning* (pp. 261–276). Routledge.
- Aronson, B., & Laughter, J. (2016). The theory and practice of culturally relevant education: A synthesis of research across content areas. *Review of Educational Research*, 86(1), 163–206.
- Batthey, D., Leyva, L. A., Williams, I., Belizario, V. A., Greco, R., & Shah, R. (2018). Racial (Mis)Match in Middle School Mathematics Classrooms: Relational Interactions as a Racialized Mechanism. *Harvard Educational Review*, 88, 455–482.
- Benson, T. A., & Fiarman, S. E. (2021). *Unconscious bias in schools: A developmental approach to exploring race and racism* (rev. ed.). Harvard Education Press.
- Benson, P. L., Scales, P. C., Hamilton, S. F., & Sesma, A. (2006). Positive youth development: Theory, research and application. In R. M. Lerner (Ed.), *Theoretical models of human development. Volume 1*.
- Bonilla, S., Dee, T. S., & Penner, E. K. (2021). Ethnic studies increases longer-run academic engagement and attainment. *Proceedings of the National Academy of Sciences*, 118(37), 1–10. <https://doi.org/10.1073/pnas.2026386118>.
- Castro-Schilo, L., & Grimm, K. J. (2018). Using residualized change versus difference scores for longitudinal research. *Journal of Social and Personal Relationships*, 35(1), 32–58. <https://doi.org/10.1177/0265407517718387>.
- Ceccon, C., Schachner, M. K., Lionetti, F., Pastore, M., Umaña-Taylor, A. J., & Moscardino, U. (2023). Efficacy of a Cultural Adaptation of the Identity Project Intervention Among Adolescents Attending Multiethnic Classrooms in Italy: A Randomized Controlled Trial. *Child Development*, 94, 1162–1180. <https://doi.org/10.1111/cdev.13944>.
- Choi, Y. (2013). Teaching Social Studies for Newcomer English Language Learners: Toward Culturally Relevant Pedagogy. *Multicultural Perspectives*, 15, 12–18.
- Christianakis, M. (2011). Hybrid texts: Fifth graders, rap music, and writing. *Urban Education*, 46, 1131–1168. <https://doi.org/10.1177/0042085911400326>.
- Delale-O'Connor, L., & Graham, D. L. (2018). Teachers' talk about race and caregiver support: "You can NEVER be too sure about parents". *Urban Education*, 54, 499–534. <https://doi.org/10.1177/0042085918806941>.
- Douglass, S., & Umaña-Taylor, A. J. (2015). A brief form of the Ethnic Identity Scale: Development and empirical validation. *Identity: An International Journal of Theory and Research*, 15, 48–65. <https://doi.org/10.1080/15283488.2014.989442>.
- Egalite, A. J., & Kisida, B. (2018). The Effects of Teacher Match on Students' Academic Perceptions and Attitudes. *Educational Evaluation and Policy Analysis*, 40, 59–81.
- Erikson, E. H. (1968). *Youth and crisis*. WW Norton & Company.
- Gay, G. (2000). *Culturally Responsive Teaching: Theory, Research, and Practice*. New York: Teachers College Press.
- Huguley, J. P., Wang, M. T., Vasquez, A. C., & Guo, J. (2019). Parental ethnic-racial socialization practices and the construction of children of color's ethnic-racial identity: A research synthesis and meta-analysis. *Psychological Bulletin*, 145, 437–458.
- Humes, K. R., Jones, N. A., & Ramirez, R. R. (2011). *Overview of Race and Hispanic Origin: 2010*. 2010 Census Briefs, issued March, 20111, C2010BR-02. Washington, DC.
- Karataş, S., Eckstein, K., Noack, P., Rubini, M., & Crocetti, E. (2023). Meeting in school: Cultural diversity approaches of teachers and intergroup contact among ethnic minority and majority adolescents. *Child Development*, 94(1), 237–253.
- Ladson-Billings, G. (1994). *The Dreamkeepers: Successful Teachers of African American Children* (1st ed.). Jossey-Bass Publishers.
- Marsiglia, F. F., Dustman, P., Harthun, M., Ritland, C., & Umaña-Taylor, A. J. (2017). Community based effectiveness trials as a means to disseminate evidence based and culturally responsive behavioral health interventions. *Health & Social Work*, 42, 108–116.
- Massey, D. S. (2014). The Racialization of Latinos in the United States. In S. M. Bucerius & M. Tonry (Eds.), *The Oxford Handbook of Ethnicity, Crime, and Immigration* (online ed.). Oxford Academic. <https://doi.org/10.1093/oxfordhb/9780199859016.013.002>.
- McArdle, J. J., & Hamagami, F. (2001). Latent difference score structural models for linear dynamic analyses with incomplete longitudinal data. In L. M. Collins, & A. G. Sayer (Eds.), *New methods for the analysis of change* (pp. 139–175). American Psychological Association. <https://doi.org/10.1037/10409-005>.
- Meeus, W. (2011). The study of adolescent identity formation 2000–2010: A review of longitudinal research. *Journal of Research on Adolescence*, 21, 75–94. <https://doi.org/10.1111/j.1532-7795.2010.00716.x>.
- Moffitt, U., & Rogers, L. O. (2022). Studying ethnic-racial identity among White youth: White supremacy as a developmental context. *Journal of Research on Adolescence*, 32, 815–828.
- Morris, S. B., & DeShon, R. P. (2002). Combining effect size estimates in meta-analysis with repeated measures and independent-groups designs. *Psychological Methods*, 7(1), 105–125. <https://doi.org/10.1037/1082-989X.7.1.105>.
- Muthén, L. K., & Muthén, B. O. (1998–2017). *Mplus User's Guide*. Eighth Edition. Los Angeles, CA: Muthén & Muthén.
- Nojan, S. (2020). Why ethnic studies? Building critical consciousness among middle school students. *Middle School Journal* 51(2), 25–35. <https://doi.org/10.1080/00940771.2019.1709259>.
- Okonofua, J. A., Walton, G. M., & Eberhardt, J. L. (2016). A vicious cycle: A social-psychological account of extreme racial disparities in school discipline. *Perspectives on Psychological Science*, 11, 381–398. <https://doi.org/10.1177/1745691616635592>.
- Paris, D. (2021). Culturally sustaining pedagogies and our futures. *The Educational Forum*, 85, 364–376. <https://doi.org/10.1080/00131725.2021.1957634>.
- PEN America (2022). PEN America Index of Educational Gag Orders. *PEN.org*. Retrieved from https://docs.google.com/spreadsheets/d/1Tj5WQVBmB6SQg-zP_M8uZsQQGH09TxBY73v23pyr0/edit#gid=1505554870.
- Rodriguez, J. L., Bustamante Jones, E., Peng, V. O., & Park, C. D. (2004). Promoting Academic Achievement and Identity Development among Diverse High School Students. *High School Journal*, 87(3), 44–53.
- Satterthwaite-Freiman, M., Umana-Taylor, A. J., Martinez-Fuentes, S., & Sladek, M. R. (2024). Examining teachers' culturally sustaining pedagogical practices during and after their implementation of an ethnic-racial identity curriculum. *Manuscript in progress*.
- Schachner, M. K., Holscher, S., Moscardino, U., Ceccon, C., Juang, L., & Pastore, M. (2023). Adolescent cultural identity development in context: The dynamic interplay of the Identity Project with classroom cultural diversity climate in Italy and Germany. *Manuscript submitted for publication*.
- Sladek, M. R., Umaña-Taylor, A. J., Wantchekon, K. A., McDermott, E. R., & Updegraff, K. A. (2021). Contextual moderators of a school-based ethnic-racial identity intervention: The roles of family ethnic socialization and ethnic-racial background. *Prevention Science*, 22, 378–385. <https://doi.org/10.1007/s1121-020-01166-8>.
- Sladek, M. R., Umaña-Taylor, A. J., McDermott, E. R., Rivas-Drake, D., & Martinez-Fuentes, S. (2020). Testing invariance of ethnic-racial discrimination and identity measures for adolescents across ethnic-racial groups and contexts. *Psychological Assessment*, 32, 509–526. <https://doi.org/10.1037/pas0000805>.
- Steinberg, L. (2005). Cognitive and affective development in adolescence. *Trends in cognitive sciences*, 9(2), 69–74.
- Sterrett, E. M., Jones, D. J., McKee, L. G., & Kincaid, C. (2011). Supportive Non-Parental Adults and Adolescent Psychosocial

- Functioning: Using Social Support as a Theoretical Framework. *American Journal of Community Psychology*, 48, 284–295. <https://doi.org/10.1007/s10464-011-9429-y>.
- Sleeter, C., & Zavala, M. (2020). *Transformative ethnic studies in schools: Curriculum, Pedagogy, and Research*. New York: Teachers College Press.
- Tajfel, H., & Turner, J. C. (1986). The social identity theory of intergroup behaviour. In Worchel, S., & Austin, W. G. (Eds.), *The psychology of intergroup relations* (Vol. 2). New York, NY: Nelson Hall.
- Umaña-Taylor, A. J. (2011). Ethnic Identity. In Schwartz, S. J., Luyckx, K., & Vignoles, V. L. (Eds.) *Handbook of Identity Theory and Research* (pp. 791–809). New York: Springer.
- Umaña-Taylor, A. J. (2018). Intervening in cultural development: the case of ethnic-racial identity. *Development & Psychopathology*, 30, 1907–1922. <https://doi.org/10.1017/S0954579418000974>.
- Umaña-Taylor, A. J. (2023a). Promoting adolescent adjustment by intervening in ethnic-racial identity development: Opportunities for developmental prevention science and considerations for a global theory of change. *International Journal of Behavioral Development*, 47(4), 352–365. <https://doi.org/10.1177/01650254231162614>.
- Umaña-Taylor, A. J. (2023b). Leveraging Ethnic-Racial Identity Development in Schools to Promote Ethnic-Racial Equity. *Manuscript submitted for publication*.
- Umaña-Taylor, A. J., & Douglass, S. (2017). Developing an ethnic-racial identity intervention from a developmental perspective: Process, content, and implementation. In Cabrera, N. J., & Leyendecker, B. (Eds.), *Handbook of positive development of minority children and youth* (pp. 437–453). Springer.
- Umaña-Taylor, A. J., Douglass, S., Updegraff, K. A., & Marsiglia, F. (2018). Small-scale randomized efficacy trial of the *Identity Project*: Promoting adolescents' ethnic-racial identity exploration and resolution. *Child Development*, 89, 862–870. <https://doi.org/10.1111/cdev.12755>.
- Umaña-Taylor, A. J., Kornienko, O., Douglass Bayless, S., & Updegraff, K. A. (2018). A universal intervention program increases ethnic-racial identity exploration and resolution to predict adolescent psychosocial functioning one year later. *Journal of Youth and Adolescence*, 47, 1–15. <https://doi.org/10.1007/s10964-017-0766-5>.
- Umaña-Taylor, A. J., Quintana, S. M., Lee, R. M., Cross, W. E., Rivas-Drake, D., & Schwartz, S. J., Ethnic and Racial Identity in the 21st Century Study Group. (2014). Ethnic and racial identity during adolescence and into young adulthood: An integrated conceptualization. *Child Development*, 85, 21–39. <https://doi.org/10.1111/cdev.12196>.
- Umaña-Taylor, A. J., & Rivas-Drake, D. (2021). Ethnic-racial identity and adolescents' positive development in the context of ethnic-racial marginalization: Unpacking risk and resilience. *Human Development*, 65(5–6), 293–310. <https://doi.org/10.1159/000519631>.
- Umaña-Taylor, A. J., Yazedjian, A., & Bámaca-Gómez, M. Y. (2004). Developing the Ethnic Identity Scale using Eriksonian and Social Identity perspectives. *Identity: An International Journal of Theory and Research*, 4, 9–38.
- Verhoeven, M., Poorthuis, A. M. G., & Volman, M. (2019). The Role of School in Adolescents' Identity Development. A Literature Review. *Educational Psychology Review*, 31, 35–63. <https://doi.org/10.1007/s10648-018-9457-3>.
- Villavicencio, A., Conlin, D., & Pagan, O. (2023). Research-practice partnerships in pursuit of racial justice in schools: Navigating a hostile sociopolitical climate. *Educational Policy*, 37, 250–275. <https://doi.org/10.1177/08959048221130353>.
- Wang, M. T., & Eccles, J. S. (2012). Social support matters: Longitudinal effects of social support on three dimensions of school engagement from middle to high school. *Child Development*, 83, 877–895.
- Waxman, O. B. (2022). Anti-‘Critical Race Theory’ Laws Are Working. Teachers Are Thinking Twice About How They Talk About Race. *Time Magazine*, June 30, 2022. Retrieved from <https://time.com/6192708/critical-race-theory-teachers-racism/>.
- Yarnell, L. M., & Bohrnstedt, G. W. (2018). Student-Teacher Racial Match and Its Association With Black Student Achievement: An Exploration Using Multilevel Structural Equation Modeling. *American Educational Research Journal*, 55, 287–324.

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